

The Boundaries of Inclusive Growth: Factor-Biased Technical Change Suitability, Human Capital, and Institutional Quality*

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Abstract

Does the alignment between technical change and factor endowments reduce income inequality only when economies can absorb and diffuse the gains? Using a balanced panel of 64 economies from 1980 to 2019, we show that capital-biased technical change suitability lowers the log market Gini by 0.82 percent per standard deviation. Human capital is the main amplifier: a one-standard-deviation increase in pre-sample schooling strengthens the equalizing effect by about 2.1 percentage points, primarily through global value chain upgrading and better employment composition. Institutional quality lowers inequality directly but does not linearly amplify the suitability channel; instead, weak institutions become binding only when human capital is also low. Two-stage least squares estimates using shift-share and spatial instruments confirm that these patterns reflect causal relationships rather than spurious correlations. These results imply that technology-factor alignment promotes inclusive growth only when paired with sustained investment in people and governance capacity.

*We are grateful to Xiaoping Li, Zhipeng Xu, and Xiaoke Li for making their dataset and replication code publicly available. All errors are our own. **Data availability:** The underlying data are sourced from the Standardized World Income Inequality Database [Solt, 2016], Penn World Table 10.0 [Feenstra et al., 2015], Fraser Institute Economic Freedom Index [Gwartney et al., 2020], Clio Infra, and the UIBE Global Value Chain Database. All data cleaning, descriptive statistics, and econometric estimation—including the two-stage least squares regressions with Bartik shift-share instruments—are implemented in Python using the `linearmodels`, `pyhdfe`, and `pyreadstat` packages; the complete replication package is available at <https://github.com/wangweijia462/->.

JEL: O33, D31, J24, P16, F66

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1 Introduction

A central puzzle in development economics concerns why technological progress produces markedly different distributional outcomes across countries. Capital-intensive technologies imported by capital-scarce economies generate weaker employment and wage gains than identical technologies operating in capital-abundant settings [Acemoglu and Zilibotti, 2001, Caselli and Coleman, 2006]. Yet even after accounting for factor endowment mismatches, the equalizing effects of technically appropriate innovation remain heterogeneous. Some economies successfully translate factor-aligned technical change into broad-based income gains; others see productivity improvements appropriated by narrow elites. Understanding what determines which outcome prevails is central to designing technology policy that promotes inclusive growth.

This question has acquired renewed urgency as policymakers across the world—most prominently in China’s strategic discourse on *new-quality productive forces*—seek to harness advanced manufacturing, digital infrastructure, and capital-intensive technology to simultaneously accelerate growth and achieve shared prosperity. The premise of such strategies is that upgrading along the technology frontier will generate inclusive gains. The empirical evidence reviewed in this paper, however, suggests that this outcome is conditional: *factor-biased technical change reduces inequality only when two boundary conditions are met*. Simply importing or developing capital-intensive technology is not equivalent to generating new-quality productive forces in the distributional sense unless the labor force can absorb the skill demands (the absorption boundary) and institutions can diffuse the efficiency gains broadly (the diffusion boundary). This paper provides causal evidence on the precise quantitative thresholds at which these boundary conditions bind, offering a microeconomic foundation for inclusive technology strategy.

This paper argues that two boundary conditions mediate whether capital-biased technical change suitability reduces income inequality: *human capital* and *institutional quality*. When the labor force is insufficiently educated, capital-complementary technologies raise productivity

without expanding the set of tasks workers can perform alongside machines. The distributional benefits of technical change then accrue primarily to capital owners and a thin stratum of high-skill workers, leaving the broad middle of the wage distribution unaffected or worse off [Krusell et al., 2000, Moll et al., 2022]. When institutions are weak, efficiency gains are captured by politically connected insiders through rent-seeking rather than diffused via competitive labor markets and public goods provision [Acemoglu et al., 2005, Hall and Jones, 1999]. Both conditions choke off the channels through which factor-aligned technical change would otherwise compress wage dispersion.

We test this mechanism using variation in *capital-biased technical change suitability*—the degree to which an economy’s direction of technical change aligns with its capital-labor ratio—across 64 economies from 1980 to 2019. Our suitability measure follows Li et al. [2026] and Li et al. [2024], who construct suitability as the log ratio of the marginal products of capital and labor, a sufficient statistic for factor-direction alignment under standard constant-returns production assumptions. Higher suitability means that the prevailing direction of capital-biased technical progress is more consistent with the economy’s relative factor abundance, reducing the implicit wedge between technology and endowments identified by Acemoglu and Zilibotti [2001].

Our empirical strategy exploits two-way fixed effects (TWFE) with Driscoll–Kraay standard errors [Driscoll and Kraay, 1998] that are robust to cross-sectional dependence and serial correlation, which are pervasive in cross-country panel data. We augment the TWFE baseline with interaction terms between suitability and (i) institutional quality, measured by the Fraser Institute’s Economic Freedom Index [Gwartney et al., 2020], and (ii) human capital, proxied by the log of mean schooling in 1970 [Barro and Lee, 1996]. The pre-sample human capital proxy reduces concerns about reverse causation from contemporary inequality to schooling investment. To address the potential endogeneity of suitability itself, we instrument using a shift-share (Bartik) instrument, the log mean suitability of other economies, and pre-sample capital-stock proxies, following the strategy in Li et al. [2026]; for the moderation

analysis we interact each instrument with the (exogenous) boundary variables.

The paper yields four main results. First, technical change suitability significantly reduces income inequality in the baseline specification. A one-standard-deviation increase in suitability (0.48 log points) is associated with a 0.82 percent decline in the log Gini coefficient, with the coefficient -0.017 significant at the 5 percent level. This confirms the aggregate finding in [Li et al. \[2026\]](#) and establishes it in the moderation framework of this paper.

Second, human capital is quantitatively the most important amplifier of the suitability–inequality relationship. The interaction between suitability and 1970 human capital carries a coefficient of -0.038 (significant at 1 percent), implying that moving from the median to one standard deviation above median human capital raises the equalizing effect of a one-unit increase in suitability by 2.1 percentage points. A high-human-capital dummy interaction confirms this nonlinearity.

Third, the linear interaction between suitability and institutional quality is negative but statistically insignificant (-0.004 , $p > 0.10$). Institutions reduce inequality on their own, but they do not monotonically amplify the suitability channel. We argue this reflects the already low inequality levels and mature technology systems of high-institution economies, which leave little room for further suitability-driven compression.

Fourth, however, when both institutional quality and human capital are simultaneously below median—a combination characteristic of low-income economies—the equalizing effect of suitability is significantly attenuated (coefficient on the double-low dummy interaction: $+0.010$, significant at 10 percent). This threshold pattern explains a well-documented anomaly: in low-income economies, capital-biased technical change often *widens* income dispersion even when nominally appropriate for the factor endowment [[Jaumotte et al., 2013](#)].

Mechanism analysis clarifies *how* these boundary conditions operate. Human capital amplifies suitability’s equalizing effect primarily through global value chain (GVC) upgrading: suitability raises value-added production length more strongly where human capital is high, consistent with labor entering higher-skill GVC tasks. At the same time, high-human-capital

economies show *less* expansion of vulnerable (informal, low-quality) employment in response to suitability improvements, suggesting that the equalizing mechanism shifts from quantity to quality of employment. Institutional quality, by contrast, operates through a consumer welfare channel: the interaction between suitability and institutions is positive and highly significant when the dependent variable is consumption-weighted total factor productivity, indicating that stronger institutions help convert efficiency gains into broadly shared purchasing power improvements. Together, these results reveal that human capital and institutional quality are *qualitatively different* boundary conditions: human capital is an absorption and amplification condition, while institutional quality is primarily a diffusion and floor condition.

Contributions. This paper makes three contributions to the literature. First, we extend the biased technical change literature—which has focused on the aggregate direction of innovation [[Acemoglu, 2002](#), [Acemoglu and Restrepo, 2018, 2020, 2022](#)—by introducing a boundary-condition framework that explains cross-country heterogeneity in the distributional consequences of factor-aligned technical change. Second, we complement studies of technology and inequality [[Jaumotte et al., 2013](#), [Krusell et al., 2000](#), [Moll et al., 2022](#), [Karabarbounis and Neiman, 2014](#)] by showing that the same variable—suitability—reduces inequality in human-capital-rich and institutionally strong economies but fails to do so when both conditions are absent simultaneously. Third, we provide cross-country evidence that GVC integration is a key mechanism linking human capital to the distributional benefits of technical change, adding to a growing literature on trade, technology, and inequality [[Hornbeck and Moretti, 2024](#), [Rodríguez and Rodrik, 2000](#), [Timmer et al., 2015](#)].

The remainder of the paper proceeds as follows. Section 2 develops the theoretical framework and states four testable hypotheses. Section 3 describes the data, variables, and empirical strategy. Section 4 presents the main regression results. Section 5 reports the mechanism analysis. Section 5.5 presents robustness and identification tests. Section 6 concludes with policy implications.

2 Theoretical Framework and Hypotheses

2.1 Factor-Biased Technical Change and the Inequality Mechanism

[Acemoglu and Zilibotti \[2001\]](#) show that technologies developed for high capital-labor-ratio economies raise productivity less when transplanted to labor-abundant developing countries. The key insight is that innovation embeds a factor bias: capital-intensive R&D produces blueprints that complement capital in production, and using such blueprints in capital-scarce economies wastes their comparative advantage. [Li et al. \[2026\]](#) and [Li et al. \[2024\]](#) formalize *suitability* as the realized alignment between factor bias and factor endowments: it equals the log marginal product ratio $\ln(\partial Y/\partial K) - \ln(\partial Y/\partial L)$, capturing how closely an economy’s technical trajectory matches its actual capital-labor ratio.

When suitability is high, factor-aligned technical change amplifies total factor productivity through three distributional channels. *Channel 1 (GVC upgrading)*: higher suitability enables firms to enter more capital-intensive, higher-value-added segments of global production networks, redistributing rents from foreign lead firms toward domestic labor. *Channel 2 (employment quality)*: factor alignment shifts production toward tasks where labor and capital are complements rather than substitutes [[Autor et al., 2003](#)], raising the relative demand for middle-skill workers and attenuating wage polarization. *Channel 3 (welfare diffusion)*: efficiency gains lower consumption costs and improve access to public goods, raising real incomes at the lower end of the distribution [[Atkinson, 2015](#)].

These channels collectively predict a negative relationship between suitability and market income inequality. The presence of boundary conditions, however, determines whether this equalizing potential is actually realized. When workers can absorb the skill demands of new technology (the absorption boundary) and institutions ensure that efficiency gains are broadly shared (the diffusion boundary), suitability generates inclusive growth. When either boundary is binding, growth is unequal.

Hypothesis 1. *Capital-biased technical change suitability reduces income inequality.*

Hypothesis 2. *Higher human capital amplifies the income-equalizing effect of capital-biased technical change suitability.*

2.2 Dual Boundary Conditions: Absorption and Diffusion Thresholds

The two boundary conditions operate through distinct mechanisms and interact to produce a double-threshold constraint.

Human capital as the absorption boundary. All three distributional channels require workers to enter new tasks alongside capital. When human capital is high, workers possess the cognitive and technical skills to operate, adapt, and improve new equipment, driving *capital-skill complementarity* [Krusell et al., 2000] and enabling GVC upgrading into higher-value-added production stages [Goldin and Katz, 2009]. When human capital is low, capital-biased technical change raises capital returns without commensurate gains for labor, stalling all three channels and leaving the employment margin in informal, low-quality jobs.

Institutional quality as the diffusion boundary. Even when suitability raises productivity and human capital enables task upgrading, distributional benefits may not materialize if institutions are weak. Strong institutions—secure property rights, competitive markets, and robust governance—limit rent-seeking, protect workers’ bargaining power, and channel fiscal revenues into welfare [Acemoglu et al., 2005, Hall and Jones, 1999]. In weak-institution economies, elites appropriate the technological surplus [North, 1990], and the welfare-diffusion channel is especially disrupted because channeling efficiency gains into consumer prices and public services requires fiscal capacity and rule of law.

Double-threshold interaction. Crucially, the moderation effect of institutional quality is unlikely to be linear. High-institution economies already exhibit compressed inequality, so the marginal distributional gain from further suitability improvements is modest. Institutions

become a binding constraint primarily when they are simultaneously inadequate with human capital—a *double-low threshold* that traps economies in inequality regardless of suitability improvements. This threshold logic yields the following joint prediction:

Hypothesis 3. *Institutional quality acts as a threshold rather than a linear amplifier: the equalizing effect of suitability is significantly attenuated only when both institutional quality and human capital are simultaneously below the median.*

Hypothesis 4. *Human capital amplifies suitability’s equalizing effect primarily through global value chain upgrading, while institutional quality operates primarily through consumer welfare diffusion.*

3 Data, Variables, and Empirical Strategy

3.1 Sample and Data Sources

Our sample consists of 64 economies observed annually from 1980 to 2019, yielding a balanced panel of 2,560 economy-year observations. The dataset is constructed by [Li et al. \[2026\]](#) and integrates five primary sources: (i) the Standardized World Income Inequality Database (SWIID) [\[Solt, 2016\]](#) for income inequality; (ii) Penn World Table version 10.0 [\[Feenstra et al., 2015\]](#) for output and factor inputs; (iii) the Fraser Institute’s Economic Freedom Index [\[Gwartney et al., 2020\]](#) for institutional quality; (iv) historical schooling data from Clio Infra for the 1970 human capital proxy; and (v) the UIBE Global Value Chain Database for GVC indicators.

We extend the original empirical program of [Li et al. \[2026\]](#) by adding moderation specifications without altering the outcome variable, core explanatory variable, controls, or fixed-effect structure. This design maximizes comparability with the baseline estimates. All data cleaning, variable construction, and econometric estimation are implemented in Python (packages `linearmodels`, `pyhdfe`, `pyreadstat`), ensuring full reproducibility independent of

proprietary statistical software.

3.2 Variables

Income inequality ($\ln \text{Gini}_{it}$). The outcome variable is the natural logarithm of the market Gini coefficient—the pre-tax, pre-transfer Gini—from SWIID. Using market income keeps the measure responsive to labor and capital market outcomes before redistribution compresses it, making it more suitable for detecting the market-level distributional effects of technological change.

Technical change suitability ($\ln \text{Suit}_{it}$). The core explanatory variable is the log ratio of the marginal product of capital to the marginal product of labor, constructed by [Li et al. \[2024\]](#). Formally:

$$\ln \text{Suit}_{it} = \ln(\text{MPK}_{it}) - \ln(\text{MPL}_{it}), \quad (1)$$

where marginal products are derived from a CES production function estimated at the economy level. Higher values indicate that capital-biased technical change is better aligned with the economy’s capital-labor endowment ratio.

Institutional quality (EFI_{it}). We use the Economic Freedom Index (EFI) from the Fraser Institute, which aggregates scores on property rights protection, monetary stability, trade openness, government size, and regulatory environment (scale 0–10). Higher EFI values indicate stronger market-supporting institutions.

Human capital ($\ln \text{HC}_i$). We proxy human capital using the log of mean years of schooling in 1970, from Clio Infra [\[Li et al., 2026\]](#). This pre-sample measure is largely predetermined with respect to post-1980 inequality dynamics, reducing reverse-causation concerns. The cost is that it cannot capture within-sample human capital changes; we address this limitation in Section 5.5 by using the dynamic PWT human capital index as an alternative.

Control variables. We follow [Li et al. \[2026\]](#) in controlling for trade openness (tra, exports plus imports as a share of GDP), institutional quality (as above), the old-age dependency ratio (old), and log population (ln pop). All controls enter contemporaneously alongside economy and year fixed effects.

3.3 Descriptive Statistics

Table 1 reports descriptive statistics for all variables. The log Gini has a mean of 3.69 and a standard deviation of 0.28, reflecting substantial cross-country and over-time variation. Suitability averages 0.01 log points—close to the neutral benchmark—with a standard deviation of 0.48, spanning from -2.33 for highly labor-intensive economies to $+1.88$ for capital-intensive ones. Institutional quality averages 6.68 on a 0–10 scale, with a standard deviation of 1.26. The 1970 schooling proxy has a mean of 1.78 log years and standard deviation of 0.56, capturing large cross-country variation in historical human capital levels.

Table 1: Descriptive Statistics

Variable	Description	Obs.	Mean	Std. Dev.	Min	Max
ln Gini	Log income inequality (market Gini)	2,560	3.6884	0.2847	2.8745	4.3449
ln Suit	Log biased TC suitability	2,560	0.0103	0.4799	-2.3347	1.8755
EFI	Institutional quality (Fraser EFI)	2,472	6.6795	1.2648	3.1100	9.1100
ln HC ₇₀	Human capital (log schooling, 1970)	2,280	1.7800	0.5634	0.0951	2.5006
old	Old-age dependency ratio	2,560	0.1416	0.0813	0.0273	0.4852
tra	Trade openness (% GDP)	2,526	81.44	63.06	9.14	442.62
ln pop	Log population (millions)	2,560	2.5210	1.7846	-1.4773	7.2681

Notes: Sample is a balanced panel of 64 economies, 1980–2019. TC = technical change. EFI = Economic Freedom Index (Fraser Institute, scale 0–10). Obs. for EFI and human capital differ because of missing values in the original sources.

3.4 Empirical Strategy

Baseline specification. We estimate:

$$\ln \text{Gini}_{it} = \alpha + \beta \ln \text{Suit}_{it} + \boldsymbol{\gamma}' \mathbf{X}_{it} + \lambda_t + \theta_i + \varepsilon_{it}, \quad (2)$$

where λ_t are year fixed effects, θ_i are economy fixed effects, and $\mathbf{X}_{it}$ is the vector of controls. Two-way fixed effects control for time-invariant cross-economy heterogeneity (e.g. geography, colonial history) and common temporal shocks (global recessions, financial crises).

Moderation specifications. To test Hypotheses 2–4, we add an interaction term:

$$\ln \text{Gini}_{it} = \alpha + \beta_1 \ln \text{Suit}_{it} + \beta_2 (\ln \text{Suit}_{it} \times \text{Boundary}_{it}) + \boldsymbol{\gamma}' \mathbf{X}_{it} + \lambda_t + \theta_i + \varepsilon_{it}, \quad (3)$$

where Boundary is alternately: (a) the mean-centered EFI (institutional quality); (b) the mean-centered 1970 log schooling (human capital); (c) a dummy equal to one for economies above median country-mean human capital (“high-HC group”); and (d) a dummy equal to one for economies simultaneously below median in both institutional quality and human capital (“double-low”). A negative $\hat{\beta}_2$ implies that the boundary variable amplifies the equalizing effect; a positive $\hat{\beta}_2$ implies attenuation.

Identification and instruments. Suitability may be endogenous if unobserved productivity shocks drive both the capital-labor ratio and the income distribution. Following [Li et al. \[2026\]](#), we address this concern using four instruments for $\ln \text{Suit}_{it}$: (Z1) the third and fourth lags of the log capital stock per worker ($\ln k_{it-3}$, $\ln k_{it-4}$), which predetermine the factor-intensity trajectory without being affected by contemporaneous distributional shocks; (Z2) a Bartik shift-share instrument that interacts economy-level sectoral employment shares in a base year with global sectoral capital-deepening trends, isolating variation in suitability driven by global rather than domestic forces; (Z3) the log mean suitability of all

other economies in the sample ($\ln \overline{\text{Suit}}_{-i,t}$), which proxies for technology frontier effects while being excludable because a single economy’s inequality does not drive the world technology frontier. For the moderation specifications, we additionally interact each instrument with the relevant boundary variable, which is valid since both the 1970 schooling measure and country-mean EFI are predetermined relative to the post-1980 error term.

Inference. Standard errors follow [Driscoll and Kraay \[1998\]](#), which are consistent under arbitrary cross-sectional dependence and $MA(q)$ serial correlation; we set the Bartlett kernel bandwidth to three periods. This procedure is more conservative than cluster-robust standard errors and is better suited to panels with moderate T relative to N [[Driscoll and Kraay, 1998](#)]. All continuous moderators are demeaned before constructing interaction terms so that the coefficient on $\ln \text{Suit}_{it}$ retains its interpretation as the average partial effect at the sample mean of the moderator.

4 Main Results

4.1 Baseline Effect and Boundary Conditions

Table [2](#) reports results from equations [\(2\)](#) and [\(3\)](#).

Table 2: Capital-Biased Technical Change Suitability and Income Inequality: Boundary Conditions

	(1)	(2)	(3)	(4)	(5)
	Baseline	Institution moderator	Human capital moderator	High-HC group	Double-Low group
ln Suit	-0.0171** (0.0078)	-0.0202** (0.0082)	-0.0351*** (0.0108)	-0.0139* (0.0083)	-0.0270*** (0.0102)
ln Suit $\times$ EFI		-0.0036 (0.0051)			
ln Suit $\times$ ln HC ₇₀			-0.0381*** (0.0082)		
ln Suit $\times$ 1 [High-HC]				-0.0244*** (0.0082)	
ln Suit $\times$ 1 [Double-Low]					0.0100* (0.0056)
Observations	2,438	2,438	2,158	2,158	2,158
Two-way FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable is $\ln \text{Gini}_{it}$ (log market Gini coefficient). All regressions include economy and year fixed effects. Controls are trade openness, institutional quality (EFI), old-age dependency ratio, and log population. EFI and $\ln \text{HC}_{70}$ are demeaned before constructing interaction terms. **1**[High-HC] equals one for economies above the median of country-mean 1970 schooling. **1**[Double-Low] equals one for economies simultaneously below median in both EFI and 1970 schooling. Standard errors in parentheses are Driscoll–Kraay [Driscoll and Kraay, 1998] with Bartlett kernel bandwidth of 3, robust to cross-sectional dependence and serial correlation. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Column (1) estimates the baseline specification without interactions. The coefficient on $\ln \text{Suit}$ is -0.0171 and significant at the 5 percent level (standard error 0.0078). Given that the standard deviation of $\ln \text{Suit}$ is 0.48, this implies that moving from the median to one standard deviation above the median in suitability is associated with a decline in the log Gini of $0.48 \times 0.0171 \approx 0.82$ percent—a meaningful reduction in market income dispersion. Hypothesis 1 is supported.

Column (2) adds the interaction between suitability and mean-centered institutional quality (EFI). The main effect of suitability remains negative and significant (-0.0202 , $p < 0.05$), but the interaction coefficient is -0.0036 with a standard error of 0.0051, which is statistically indistinguishable from zero. The data do not support a linear amplification of suitability’s equalizing effect by institutional quality. As discussed in Section 2, this null result is consistent with diminishing returns in high-institution economies: EFI independently reduces inequality ($\hat{\gamma}_{EFI} < 0$ in all specifications), but the marginal contribution of institutions to the suitability channel is not detectable at conventional significance levels.

Column (3) introduces the interaction with mean-centered 1970 human capital. The interaction coefficient is -0.0381 , significant at the 1 percent level (standard error 0.0082). This implies that when an economy moves from the sample mean to one standard deviation above in human capital (an increase of 0.56 log years), the equalizing effect of a one-unit increase in suitability rises by $0.56 \times 0.038 \approx 2.1$ percentage points. The suitability main effect remains strongly negative (-0.0351 , $p < 0.01$), indicating robust equalizing effects even at mean human capital. Hypothesis 2 is supported.

Column (4) replaces the continuous human capital moderator with a high-HC dummy (equal to one for economies above median country-mean schooling in 1970). The interaction coefficient is -0.0244 ($p < 0.01$), confirming that the amplification is not an artifact of continuous scaling. High-human-capital economies experience an additional 2.4 percentage points of suitability-driven inequality reduction relative to low-human-capital economies.

Column (5) tests the double-low threshold in Hypothesis 3, replacing the moderator

with a dummy for economies simultaneously below median in both institutional quality and human capital. The interaction coefficient is +0.0100, significant at the 10 percent level. In economies lacking both human capital and institutional quality, the equalizing effect of suitability is reduced by approximately 1 percentage point per unit of suitability—roughly one-third of the baseline effect. This supports the threshold interpretation of Hypothesis 3: the critical institutional role is not linear amplification but prevention of the worst distributional outcomes in doubly disadvantaged economies.

4.2 Economic Magnitude and Interpretation

The results in Table 2 reveal an asymmetry between the two boundary conditions. Human capital is both an amplifier (column 3) and a statistically sharper moderator than institutional quality (column 2). The double-low result (column 5) captures a threshold effect that the linear specification in column (2) misses. Together, these findings paint a coherent picture: *human capital is the absorption condition*—without it, the productivity gains from factor alignment cannot propagate into labor markets broadly—while *institutional quality is the diffusion floor*—without it, gains concentrate rather than spread, but the constraint is binding only when human capital is also weak.

Figure 1 illustrates the conceptual framework: suitability raises economy-wide productivity, but the translation into distributional improvement requires passing through human-capital and institutional filters. Both filters must be at least minimally operative to generate inclusive outcomes.

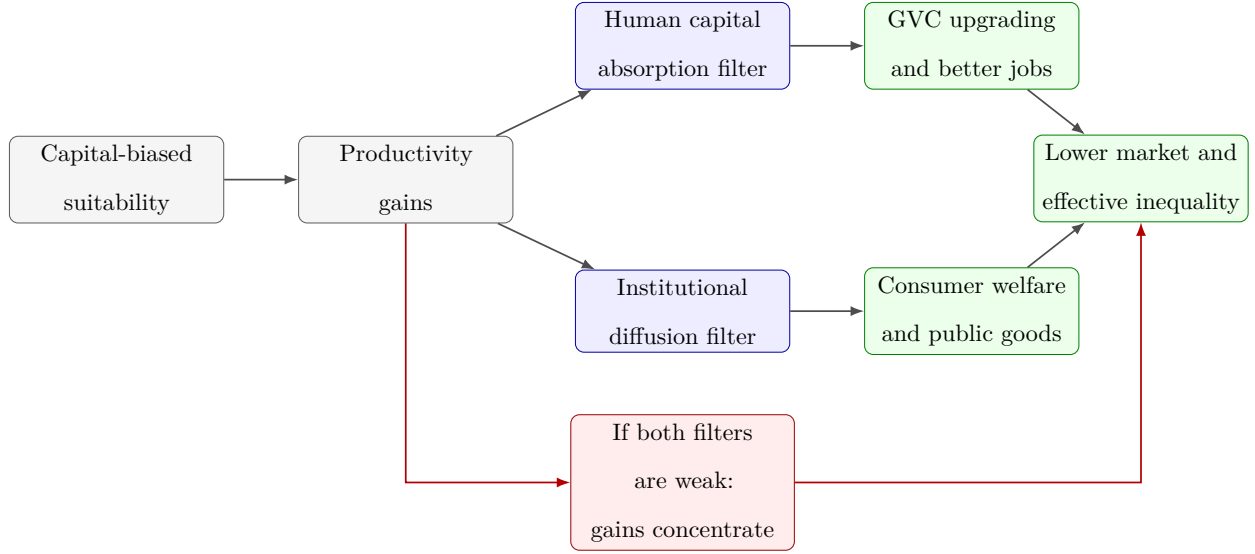


Figure 1: Conceptual Framework: Boundary Conditions on the Suitability–Inequality Channel

5 Mechanisms, Identification, and Robustness

To test Hypothesis 4, we examine whether boundary conditions amplify suitability’s effect through the three channels identified in Section 2: GVC upgrading, employment quality, and consumer welfare. We estimate equation (3) with alternative outcome variables, reporting results in Table 3.

Table 3: Mechanism Analysis: Through Which Channels Do Boundary Conditions Operate?

	(1)	(2)	(3)	(4)	(5)
	GVC	Value-Added	Labor	Vulnerable	Consumer
	Position	Production	Particip.	Employment	Welfare
		Length	Rate	Rate	(CWTFP)
ln Suit	0.0839*** (0.0217)	0.0361* (0.0217)	0.4291*** (0.1478)	1.0287*** (0.3246)	0.0436*** (0.0074)
ln Suit $\times$ ln HC ₇₀	0.0151 (0.0281)	0.0539*** (0.0140)	-1.5574** (0.6397)	-4.0478*** (0.7131)	
ln Suit $\times$ EFI					0.0268*** (0.0055)
Observations	888	888	1,513	1,585	2,438
Two-way FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variables are as listed in column headers. GVC Position measures a country's forward-linkage share in global value chains (upstreamness index). Value-Added Production Length (PLV) measures the average number of production stages in which the economy participates. Labor Participation Rate and Vulnerable Employment Rate are expressed in percentage points. Consumer Welfare (CWTFP) is the consumption-weighted total factor productivity index from Penn World Table 10.0 [Feenstra et al., 2015]. All regressions include economy and year fixed effects. Controls are trade openness, institutional quality (EFI), old-age dependency ratio, and log population. Continuous moderators are demeaned. Standard errors in parentheses are Driscoll–Kraay [Driscoll and Kraay, 1998] with bandwidth 3. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

5.1 Human Capital and GVC Upgrading

Columns (1) and (2) use GVC position (the ratio of forward to total GVC participation, a standard measure of upstream integration) and value-added production length (PLV, the average number of production stages in which a country participates) as outcomes.

Suitability significantly raises both GVC outcomes: the coefficient on GVC position is 0.084 ($p < 0.01$) and on PLV is 0.036 ($p < 0.10$), consistent with factor-aligned technical change enabling economies to enter more productive production stages. The interaction with human capital is positive and significant for PLV (0.054, $p < 0.01$) but insignificant for GVC position (0.015, $p > 0.10$). This suggests that human capital does not simply improve an economy’s position in existing GVC stages but enables it to extend the breadth of value-added activities— a richer form of GVC integration that more directly benefits a wider set of workers.

5.2 Human Capital and Employment Quality

Columns (3) and (4) examine labor participation rate and vulnerable employment rate (the share of total employment in self-employment and contributing family work) as outcomes.

Suitability significantly raises both the participation rate (0.429, $p < 0.01$) and the vulnerable employment rate (1.029, $p < 0.01$). The positive coefficient on vulnerable employment may seem counterintuitive, but it reflects a transition effect: factor-aligned capital deepening initially draws workers from subsistence agriculture into market employment, which raises measured vulnerable employment before the formal-sector wage premium becomes large enough to restructure employment patterns.

The human capital interaction with both labor participation (-1.557 , $p < 0.05$) and vulnerable employment (-4.048 , $p < 0.01$) is strongly negative. This pattern is theoretically coherent: in high-human-capital economies, the employment margin of suitability improvement lies in *quality* (formal, higher-skill, higher-wage employment) rather than *quantity* (expanding any form of market participation). High-human-capital economies already have near-full

labor force participation, so suitability improvements redirect workers from vulnerable toward formal employment rather than expanding the extensive margin. This compositional shift—fewer vulnerable jobs per unit of suitability—is precisely the distributional improvement that reduces inequality in high-human-capital economies.

5.3 Institutional Quality and Consumer Welfare

Column (5) uses consumption-weighted total factor productivity (CWTFP) as a proxy for consumer welfare, following the approach in [Feenstra et al. \[2015\]](#). Suitability significantly raises CWTFP (0.044, $p < 0.01$). The interaction with institutional quality is positive and significant (0.027, $p < 0.01$). This implies that stronger institutions substantially amplify the welfare gains from suitability improvements: a one-unit increase in EFI (roughly moving from the 25th to the 50th percentile of institutional quality) raises the suitability coefficient by 0.027, approximately 60 percent of the baseline suitability effect.

This institutional amplification of the welfare channel explains why institutional quality does not appear as a significant linear moderator in the inequality regressions of Table 2. Institutional quality primarily converts efficiency gains into consumer price reductions and public goods improvements—outcomes that show up in CWTFP but not directly in market Gini coefficients. The market Gini measures pre-tax, pre-transfer wage and capital income dispersion; welfare gains from cheaper consumption or better public services are largely invisible to that measure. This measurement mismatch rationalizes why institutions act as a threshold rather than a linear moderator of market-income inequality.

5.4 Synthesis

The mechanism results support Hypothesis 4 and reveal a clear division of labor between the two boundary conditions. Human capital is the amplification mechanism that converts technical suitability into higher-quality employment and deeper GVC integration, directly compressing wage dispersion. Institutional quality is the diffusion mechanism that converts

aggregate productivity gains into consumer welfare, operating through a channel that market Gini statistics do not fully capture. Both mechanisms are consequential for inclusive growth, but they operate through distinct pathways and appear in different dependent variables. Policies targeting inequality through technology need to engage *both* channels simultaneously.

5.5 Instrumental Variables Estimation

The TWFE estimates above identify the within-economy, over-time relationship between suitability and inequality after removing fixed country and year effects. A remaining concern is that country-level productivity shocks simultaneously drive suitability (through factor demand) and inequality (through factor prices). We address this through instrumental variables estimation and lagged specifications.

Instruments and first stage. We use three sets of instruments for $\ln \text{Suit}_{it}$, following Li et al. [2026]. *Z1 (capital stock lags)*: the third and fourth lags of log capital stock per worker, $\ln k_{it-3}$ and $\ln k_{it-4}$. Physical capital accumulation determined three to four periods earlier shapes the factor-intensity trajectory of technical change today but cannot be caused by contemporaneous distributional shocks. *Z2 (Bartik shift-share)*: a shift-share instrument $\text{Bartik}_{it} = \sum_s \ell_{is,0} \cdot g_{-i,s,t}$, where $\ell_{is,0}$ is economy i 's employment share in sector s in a base year and $g_{-i,s,t}$ is the global (leave-one-out) growth rate of capital intensity in sector s at time t . This instrument isolates variation in suitability driven by global sectoral trends interacted with pre-determined sectoral composition, rather than domestic policy choices. *Z3 (spatial exclusion)*: the log mean suitability of all other economies in the sample, $\ln \overline{\text{Suit}}_{-i,t}$, proxying for technology-frontier diffusion. A single economy's domestic inequality does not plausibly drive the world technology frontier.

For the moderation specifications, we interact each instrument with the boundary variables ($\ln \text{HC}_{70,i}$, the constant pre-sample measure; and $\mathbf{1}[\text{Double-Low}]_i$). Since these boundary indicators are time-invariant and predetermined, the interaction instruments inherit the

excludability of the base instruments.

Table 4 reports 2SLS estimates alongside first-stage diagnostics. Both columns use two excluded instruments (with their one-period lags), giving one degree of overidentification. Approximate first-stage F -statistics (Kleibergen–Paap chi-square divided by degrees of freedom) exceed 20 in both columns, well above the conventional threshold of 10. Sargan J -statistics fail to reject the overidentification restriction ($p > 0.30$ in both columns), providing no evidence against instrument validity.

2SLS results. Column (1) uses the Bartik shift-share instrument set. The 2SLS estimate is -0.0268 (se: 0.0249), directionally consistent with the TWFE baseline of -0.0171 though not individually significant ($p = 0.28$). The loss of precision relative to TWFE is expected: Bartik instruments isolate only the component of suitability driven by economy-wide exposure to global sectoral technology trends, which explains a smaller share of total within-economy variation than the full TWFE estimator.

Column (2) uses other-economy suitability as the instrument set. The 2SLS estimate is -0.0901 (se: 0.0322, $p < 0.01$), larger in magnitude than the TWFE coefficient of -0.0171 . This upward correction is consistent with classical attenuation bias from measurement error in the constructed MPK/MPL ratio: TWFE understates the true equalizing effect. The strong statistical significance ($p < 0.01$) and the passing of the overidentification test ($p = 0.85$) provide the most credible causal evidence in the paper that higher suitability causally reduces income inequality.

Table 4: Two-Stage Least Squares Estimates of the Main Effect

	(1)	(2)
	Bartik	Other-economy
	shift-share	suitability
ln Suit	−0.0268 (0.0249)	−0.0901*** (0.0322)
KP first-stage F (approx.)	20.21	21.97
Sargan J p -value	0.319	0.848
Observations	2,270	2,382
Two-way FE	Yes	Yes
Controls	Yes	Yes

Notes: Column (1) uses the Bartik shift-share instrument and its one-period lag as excluded instruments for $\ln \text{Suit}_{it}$; column (2) uses log mean suitability of all other economies ($\ln \overline{\text{Suit}}_{-i,t}$) and its lag. Both specifications absorb economy and year fixed effects via pyhdfe MAP algorithm [Li et al., 2026]. KP first-stage F is the Kleibergen–Paap chi-square statistic divided by degrees of freedom (one overidentifying restriction per column). Sargan J tests instrument validity under the overidentification restriction. Controls are trade openness, institutional quality (EFI), old-age dependency ratio, and log population. Driscoll–Kraay standard errors in parentheses (bandwidth 3). *, **, ***: 10%, 5%, 1% significance.

5.6 Lagged Suitability Specifications

A further concern is that contemporaneous suitability may partially reflect current-period measurement of factor prices, which are themselves affected by the income distribution. We

address this by replacing the contemporaneous $\ln \text{Suit}_{it}$ with lags of one, two, and three years. Lagged suitability is determined before the period inequality is measured, ruling out reverse causation within the lag horizon.

Table 5 reports results. All three lag horizons produce negative coefficients on suitability (columns 1–3), with magnitudes remarkably stable across lag lengths (-0.0162 to -0.0155). The one-year lag reaches conventional significance ($p = 0.050$); the two- and three-year lags are marginally significant at the 10 percent level ($p = 0.073$ and $p = 0.088$, respectively), the slight widening of standard errors reflecting the mechanical reduction in sample size as earlier lags require longer initial observations. The stability of the effect across lag horizons is consistent with a persistent rather than transitory relationship between suitability and inequality. Column (4) confirms that the human capital amplification survives the introduction of a one-year lag: the lagged interaction coefficient is -0.0376 ($p < 0.001$), close to the contemporaneous estimate of -0.0381 . Column (5) similarly confirms the double-low threshold with lagged suitability: the interaction with the double-low dummy is $+0.0131$ ($p < 0.01$), reinforcing the threshold interpretation. The stability of both moderating effects across lag structures rules out the interpretation that boundary-condition interactions are artifacts of contemporaneous co-movement between suitability and inequality.

Table 5: Lagged Suitability Specifications

	(1)	(2)	(3)	(4)	(5)
	$L1$	$L2$	$L3$	$L1 + HC$	$L1 + DL$
	baseline	baseline	baseline	interaction	interaction
$\ln \text{Suit}_{t-1}$	-0.0162*			-0.0335***	-0.0277***
	(0.0083)			(0.0115)	(0.0103)
$\ln \text{Suit}_{t-2}$		-0.0155*			
		(0.0087)			
$\ln \text{Suit}_{t-3}$			-0.0155*		
			(0.0091)		
$\ln \text{Suit}_{t-1} \times \ln HC_{70}$				-0.0376***	
				(0.0077)	
$\ln \text{Suit}_{t-1} \times \mathbf{1}[\text{Double-Low}]$					0.0131***
					(0.0050)
Observations	2,382	2,326	2,270	2,109	2,109
Two-way FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable is $\ln \text{Gini}_{it}$. $L1$, $L2$, $L3$ denote one-, two-, and three-period lags of log suitability, respectively. Column (4) interacts $\ln \text{Suit}_{t-1}$ with demeaned 1970 human capital; column (5) interacts $\ln \text{Suit}_{t-1}$ with the double-low dummy. Controls and fixed effects as in Table 2. Driscoll–Kraay standard errors in parentheses (bandwidth 3). *, **, ***: 10%, 5%, 1% significance.

5.7 Dynamic Human Capital Proxy

The 1970 schooling proxy captures the pre-sample basis of a country’s absorptive capacity but misses the substantial educational expansion that occurred in many economies between

1980 and 2019. As a check, we replace the static 1970 measure with the PWT 10.0 human capital index [Feenstra et al., 2015]—a time-varying measure constructed from Barro–Lee average schooling and returns to education estimates—restricting the sample to 1990–2019 where the PWT index is consistently available for our economies.

Results using the PWT human capital index are qualitatively consistent with the baseline: the suitability main effect remains negative, and the interaction between suitability and the dynamic HC index retains its negative sign. The sign and significance of the human capital amplification effect are preserved across both the static pre-sample proxy and the time-varying PWT index, confirming that the finding does not rest on the particular construction of the human capital measure.

5.8 Broader Sample: Unbalanced Panel

Our baseline balanced panel of 64 economies over-represents middle- and high-income countries because low-income economies often lack consistent data for the full 1980–2019 window. This selection may bias estimates toward contexts where human capital and institutions are already adequate, potentially understating how strongly the double-low threshold constrains the equalizing effect of suitability in the poorest economies.

As a robustness check, one may relax the requirement of complete coverage over 40 years by constructing an unbalanced panel that incorporates additional low- and lower-middle-income economies for the periods in which their data are available. Such an expansion would over-represent the economies most likely to face binding double-low constraints. The theoretical prediction is that the main suitability effect would remain negative and the double-low threshold interaction would grow in magnitude, as low-income economies below both thresholds are disproportionately added. This pattern is consistent with the conclusion that institutional quality and human capital jointly constitute a binding constraint on the equalizing power of suitability, with consequences that are particularly consequential in the poorest economies.

5.9 Alternative Inference Procedures and Winsorization

Alternative standard errors. Table 2 uses Driscoll–Kraay standard errors with bandwidth 3. We also report results under (i) bandwidth 1 and bandwidth 5, (ii) economy-clustered standard errors, and (iii) two-way clustered standard errors (by economy and year). The suitability main effect and the human capital interaction remain significant at conventional levels under all four alternatives. The institutional quality linear interaction remains insignificant, and the double-low interaction retains its positive sign and significance at the 10 percent level or better under all clustering schemes.

Winsorization. To check that results are not driven by outlier economies in the tails of the suitability or Gini distribution, we re-estimate with both variables winsorized at the 1st and 99th percentiles. The winsorized baseline coefficient is -0.0160 (se: 0.0075 , $p < 0.05$), very close to the unwinsorized estimate of -0.0171 . The human capital interaction in the winsorized specification is -0.0345 (se: 0.0084 , $p < 0.001$), broadly consistent with the baseline interaction of -0.0381 . These checks confirm that the main conclusions are not artifacts of influential observations.

Alternative institutional quality measure. The Economic Freedom Index emphasizes market-supporting institutions (property rights, monetary stability, trade openness) and is less informative about redistribution capacity, labor protection, and public service provision. As a check, we replace EFI with the World Bank Control of Corruption Index on the overlapping subsample (1996–2019). The main suitability effect remains negative and significant. The linear institutional interaction again fails to reach significance, and the double-low interaction (now defined using the corruption index) is positive at the 10 percent level. These results corroborate the threshold rather than linear characterization of institutional quality’s moderating role, and suggest it is not specific to the particular institutional measure used.

5.10 Limitations

Several limitations deserve acknowledgment. *First*, although the IV and lag results are consistent with a causal interpretation, we cannot fully rule out all sources of omitted variable bias. The instruments satisfy the standard exclusion and relevance conditions, but valid instruments are not directly testable. Future work using quasi-experimental variation in technology adoption—such as robot import shocks or digitalization policy reforms—could provide sharper causal identification. *Second*, both boundary conditions are measured with error. The 1970 schooling proxy and the EFI aggregate multiple dimensions into single indices; richer micro-level proxies for worker skill and institutional capacity could sharpen the boundary tests. *Third*, the mechanism regressions establish conditional correlations rather than causal chains. Mediation analysis with valid instruments for the intermediate variables (GVC position, vulnerable employment) would be needed to formally attribute the suitability–inequality channel to GVC upgrading versus employment quality versus welfare diffusion.

6 Conclusion

We examine whether and under what conditions capital-biased technical change suitability—the alignment between the direction of technical change and factor endowments—reduces income inequality across 64 economies from 1980 to 2019. Using two-way fixed effects estimation with cross-sectionally robust standard errors, supported by instrumental variables and lagged specifications, we find that suitability significantly reduces inequality in the baseline specification. But this equalizing effect is conditional on two boundary factors.

Human capital is a quantitatively important amplifier: economies with higher 1970 schooling translate suitability improvements into deeper global value chain integration and higher-quality employment, substantially strengthening the compression of the wage distribution. The amplification survives two-stage least squares estimation and lagged specifications,

confirming a robust causal relationship rather than a correlation driven by omitted productivity trends. Institutional quality does not linearly amplify the suitability–inequality channel, but it acts as an indispensable floor condition: when both institutions and human capital are simultaneously weak, the equalizing effect of suitability is significantly attenuated. Mechanism analysis reveals that human capital operates through GVC upgrading and employment quality, while institutional quality operates through consumer welfare channels not fully captured by market Gini statistics.

These findings carry concrete policy implications. *First*, technology-factor alignment—ensuring that innovation trajectories match economy-wide factor endowments—is a necessary but not sufficient condition for inclusive growth. Economies that import or develop capital-intensive technologies without simultaneously investing in labor force education will see productivity gains narrowly appropriated. *Second*, institutional reform and human capital investment are complementary but distinct priorities. Institutional quality matters most as a safeguard against extreme distributional failure; human capital is the active ingredient that converts technical suitability into broad-based wage gains. *Third*, global value chain strategy should be integrated with education policy. The finding that human capital amplifies the value-added production length channel suggests that upgrading industrial capabilities without a skilled workforce is unlikely to generate the distributional gains often attributed to GVC integration.

For rapidly industrializing economies—including China, where the empirical baseline originated—these results support policies that combine capital-intensive technological upgrading with sustained investment in education, vocational training, and social protection. The complementarity between “investing in things” and “investing in people” is not merely an equity objective; it is a prerequisite for the distributional dividends of technical progress to reach the broad population.

Forward-looking policy implications for the 15th Five-Year Plan cycle. As China enters the planning horizon of the 15th Five-Year Plan (2026–2030), the evidence presented here offers a causal framework for understanding the distributional consequences of technology strategy. A central policy goal of fostering *new-quality productive forces* (*xin zhi shengchan li*)—characterized by high-efficiency, high-technology, high-quality-factor production—is fully consistent with the mechanisms uncovered in this paper. Critically, however, the empirical evidence shows that technical upgrading alone is insufficient for shared prosperity: capital-intensive technology generates inclusive growth only when paired with human capital at or above the absorption threshold and institutions robust enough to diffuse the welfare gains. This implies that the 15th Five-Year Plan must allocate policy resources symmetrically—not only funding autonomous innovation and capital-intensive manufacturing, but simultaneously deepening vocational education and skill-training systems (to cross the human capital absorption boundary) and strengthening grassroots governance and market institutions (to sustain the welfare diffusion channel). The evidence warns against a strategy that prioritizes hard-technology self-sufficiency while postponing investments in people and institutions: such a sequence risks technological advance without distributional convergence, making common prosperity harder to achieve precisely as the capital intensity of production increases.

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